

Sharpening the Supersymmetric Axion Weak Gravity Conjecture

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ABSTRACT: The Axion Weak Gravity Conjecture provides one of the most effective quantum gravity tools for constraining particle physics and cosmology, but it has long been thought of as a slightly fuzzy statement: given an axion with decay constant f there should exist an instanton of charge n and action S with $fS/|n|$ at most an order-one number in Planck units. Recent work related to axion wormholes motivated a specific order-one coefficient, $\frac{fS}{|n|} \leq \frac{\pi}{2\kappa_d} \sqrt{\frac{d-1}{d-2}}$. In this work, we verify this bound in various axion sectors across the string landscape using three complementary approaches. In the process, we derive even tighter bounds on instantons in such sectors. For example, we argue that supersymmetric instantons in 4d satisfy the stronger bound of $\frac{fS}{|n|} \leq \frac{1}{\kappa_4} \sqrt{\frac{7}{2}}$.

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1 Introduction

Among the vast landscape of quantum gravity conjectures, the axion version of the Weak Gravity Conjecture (WGC) [1] offers one of the most promising routes for bridging the gap between quantum gravity theory and experiment. Most famously, the axion WGC has been shown to place meaningful bounds on simple models of natural inflation [1], including multi-field models such as N -flation [2, 3] and axion alignment [4–9]. The conjecture also has interesting consequences for the QCD axion [10–14], dark matter [15–17], quintessence [18, 19], and early dark energy [20–22].

The axion WGC holds that in a consistent theory of quantum gravity in d dimensions, any axion $\theta \simeq \theta + 2\pi$ must have a charged instanton of instanton number n and instanton action S satisfying

$$\frac{fS}{|n|} \leq c M_{\text{Pl};d}^{\frac{d-2}{2}}, \quad (1.1)$$

where f is the axion decay constant, $M_{\text{Pl};d}$ is the reduced Planck scale in d dimensions,¹ and c is an order-one constant.

In the past decade, the precise value of the order-one coefficient c has been a source of much discussion and debate. The ordinary WGC for particles [1] was originally motivated by the requirement that non-BPS black holes can decay by emitting charged particles. Consequently, the analogous order-one coefficient is determined by the charge-to-mass ratio of an extremal black hole, and the WGC bound is simply the extremality bound. See, e.g., [23–25] for reviews.

For instantons charged under axions, however, there is no analogue of the Hawking evaporation process, so the original motivation for the WGC breaks down. Furthermore, there are multiple types of semiclassical gravitational instantons, hence there are *a priori* multiple options for the coefficient c .

The question of fixing the coefficient c has been a longstanding open problem. In the recent works [26, 27], it was argued that in a theory of Einstein gravity coupled to an axion of decay constant f , a failure of semiclassical gravity can be cured by instantons satisfying

$$\frac{fS}{|n|} \leq \frac{\pi}{2} \sqrt{\frac{d-1}{d-2}} M_{\text{Pl};d}^{\frac{d-2}{2}}. \quad (1.2)$$

While there could be other effects that also restore the validity of semiclassical gravity, this suggests the coefficient $c = \frac{\pi}{2} \sqrt{\frac{d-1}{d-2}}$ as a strong candidate for the axion WGC bound (1.1), and we will refer to this bound henceforth as the *sharpened axion WGC*.

¹Throughout this paper, we set $8\pi G_d = M_{\text{Pl};d}^{2-d} = \kappa_d^2$, with the Einstein-Hilbert term normalized to $\frac{1}{2\kappa_d^2} \int d^d x \sqrt{-g} \mathcal{R}$. When convenient, we set $M_{\text{Pl};d} = 1$.

It is saturated by *half* the action of the Giddings-Strominger wormhole [28], which was previously proposed to define the axion WGC in [7]. This is no coincidence: the argument of [26, 27] avoids pathologies associated with such wormhole solutions by requiring that a different instanton be unsuppressed at a specific imaginary distance in field space.

A sharp bound of the form (1.2) requires a clear specification of the meaning of the instanton action S . In general, instanton effects take the form $A \exp(-S)$, so there is always an ambiguity from a rescaling of the prefactor $A \mapsto \lambda A$ accompanied by an additive shift $S \mapsto S + \log \lambda$. However, in many cases of interest, there is a controlled semiclassical expansion in which S corresponds to a classical action, for example, of the Yang-Mills instanton solution or of the worldvolume of a charged object. Our interest is mostly in cases where the instanton arises from a wrapped Euclidean worldvolume, and the action is specified by the mass or tension of the charged object times the wrapped volume. In these cases, S has a clear meaning, at least at leading order in an expansion in large volumes and weak couplings. When S becomes small, it is more difficult to attach a meaning to the axion WGC, as the instanton expansion breaks down and certain states become light at least for some values of the axion field [29]. In that case, it may be that a more direct analysis of the imaginary distance bound of [26, 27] and the gravitational path integral can lead to a quantitative constraint without the need to refer to an instanton action. For now, we will restrict our attention to cases with controlled semiclassical expansions.

In this paper, we use three complementary approaches to verify and strengthen the bound (1.2) in various axion sectors in the quantum gravity landscape. First, we review the argument of [8], which translates higher-form versions of the WGC into geometric bounds on Calabi-Yau manifolds. By compactifying different string theories on these same manifolds, these geometric bounds lead to axion WGC bounds for various axion sectors. This approach has the benefit that it applies to both BPS and non-BPS instantons; it has the drawback of relying on the assumption of the WGC for higher-form gauge fields.

Next, we develop a novel approach to strengthen the axion WGC for BPS (holomorphic) instantons in 4d supergravity theories whose Kähler potential takes the form

$$\mathcal{K} = -\log(\mathcal{F}(t^i)), \tag{1.3}$$

for $\mathcal{F}(t^i)$ a homogeneous function of the moduli t^i . Such Kähler potentials are ubiquitous in string/M-theory compactifications, as we shall see. Furthermore, BPS instantons are important because they are the only instantons that can contribute to the superpotential of an $\mathcal{N} = 1$ theory, not merely the Kähler potential. From this

approach, we derive the axion WGC bound:

$$\frac{fS}{|n|} \leq \sqrt{\frac{p}{2}} M_{\text{Pl}}, \quad (1.4)$$

where p is the degree of homogeneity of $\mathcal{F}$, i.e., $\mathcal{F}(\lambda t^i) = \lambda^p \mathcal{F}(t^i)$. We argue that various Swampland conjectures imply that p is generically a positive integer satisfying $1 \leq p \leq 7$, which ensures that the sharpened axion WGC (1.2) is satisfied with room to spare. (In fact, $p = 7$ is the largest integer for which (1.2) is satisfied when (1.4) is saturated, so it is noteworthy that it is realized in examples in the landscape.) This second approach has the drawback that it applies specifically to BPS instantons in 4d; it has the advantage that it depends only on low-energy supergravity and is independent of the details of the UV completion.

For our third and final approach, we consider the so-called α -vectors² of holomorphic instantons:

$$\vec{\alpha} = \vec{\nabla} \log S, \quad (1.5)$$

where the gradient is taken with respect to the moduli of the theory. We show that as a simple consequence of holomorphy, these α -vectors satisfy

$$\|\vec{\alpha}\| = \frac{|n|}{fS}. \quad (1.6)$$

Consequently, a lower bound on the length of the α -vector translates into an upper bound on $fS/|n|$. We then derive lower bounds on lengths of instanton α -vectors in various axion sectors within the string/M-theory landscape. This approach has the drawback that it applies only to holomorphic instantons; it has the advantage that it applies to string/M-theory compactifications in any number of dimensions.

We will see that these three approaches are perfectly complementary in that the coefficients c derived by one approach perfectly match those derived by the other approaches whenever two or more approaches are simultaneously applicable. This suggests that the results of each approach may extend beyond the apparent domain of validity: for instance, bounds on BPS instantons in certain theories may extend to non-BPS instantons in those theories as well. In addition, all three of these approaches extend straightforwardly to theories with multiple axions. Where there is overlap, our results are also consistent with past explicit tabulations of fS for axions in string theory, e.g., in [16].

²The concept of these vectors originated for gradients of logarithms of masses (see, e.g., [30–39]), and has been since extended to branes or instanton actions (see, e.g., [40–45]). Here, we find it useful to use a positive sign-convention for instantonic α -vectors, differing from some of the literature.

While instantons are electrically charged under axion gauge fields, in 4d axion strings are magnetically charged under axion gauge fields. Axion strings have their own version of the weak gravity conjecture, namely that in a consistent theory of quantum gravity in d dimensions, any axion θ must have an axion string magnetically charged under θ satisfying

$$\frac{\mathcal{T}}{\tilde{g}} \leq c' M_{\text{Pl};4}, \quad (1.7)$$

where $\mathcal{T}$ is the axion string's tension, $\tilde{g}$ is the magnetic coupling of the axion string to the axion, and c' is an order-one constant. Our second and third approaches to strengthening the axion WGC can be used to find a value of c' such that all BPS axion strings charged under the axions considered in those approaches satisfy (1.7) for said value of c' ; we investigate this in §4.5 and §5.3 respectively.

The remainder of this paper is structured as follows. In §2, we introduce our conventions and review the extension of the axion WGC to theories with multiple axion fields. In §3-5, we present each of our three approaches in turn. In §3, we review the argument of [8] for the axion WGC using the duality web. In §4, we derive bounds on BPS instantons in theories with homogeneous Kähler potentials. In §5, we use the connection with α -vectors to derive bounds on BPS instantons. In §6, we comment briefly on applications of our bounds to natural axion inflation. In §7, we conclude with a summary and potential next steps.

2 Background and Conventions

In this section, we present relevant background on the Weak Gravity Conjecture (WGC) and axions, and we introduce notation that will be used in the remainder of the paper.

2.1 The Weak Gravity Conjecture

Consider a theory of real massless scalar fields ϕ^a , $a = 1, \dots, n_s$ and p -form gauge fields A_p^i , $i = 1, \dots, n$ coupled to gravity in d spacetime dimensions:

$$S = \int \sqrt{-g} d^d x \left(\frac{1}{2\kappa_d^2} \mathcal{R} - \frac{1}{2} g_{ab}(\phi) \partial_\mu \phi^a \partial^\mu \phi^b - \frac{1}{2} \tau_{ij}(\phi) F_{p+1}^i \cdot F_{p+1}^j \right). \quad (2.1)$$

Here, $F_{p+1} \cdot G_{p+1} = \frac{1}{(p+1)!} F_{\mu_1 \dots \mu_{p+1}} G^{\mu_1 \dots \mu_{p+1}}$, g_{ab} is the metric on moduli space, and $\tau_{ij}(\phi)$ is the gauge kinetic matrix, which in general depends on the moduli ϕ^a .

We work in a lattice basis, so that a $(p-1)$ -brane carries integral charge $q_i \in \mathbb{Z}^n$ under the respective gauge fields A^i . The normalized charge is then given by contracting with the inverse of the gauge kinetic matrix:

$$\|q\|^2 = q_i \tau^{ij} q_j. \quad (2.2)$$

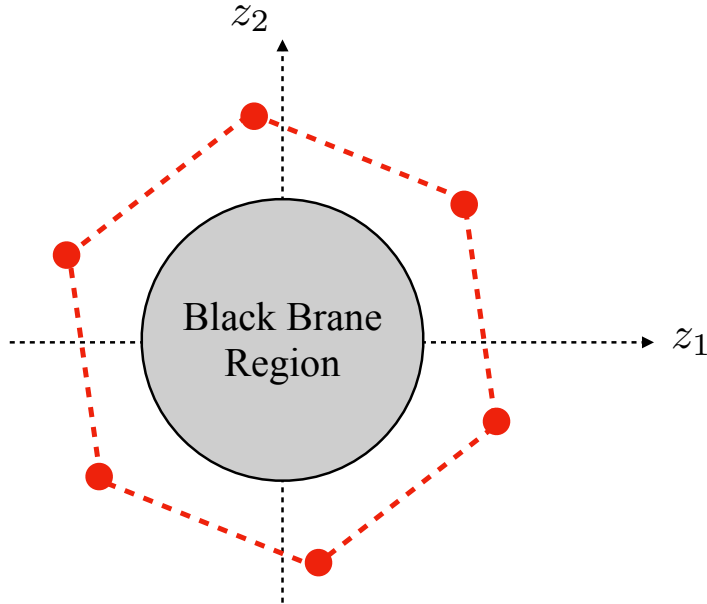


Figure 1: The convex hull condition. In theories with multiple gauge fields, the WGC is equivalent to the statement that the convex hull of the charge-to-tension vectors z_i of the charged branes (red) must contain the black brane region (shaded gray).

It is useful to define the charge-to-tension vector $z_i = q_i/(T\kappa_d)$, where T is the tension of the brane. The norm of such a z -vector is also determined by contracting with the inverse gauge kinetic matrix, $\|z\| = z_i \tau^{ij} z_j$.

Given such a theory, the WGC requires a superextremal $(p-1)$ -brane state in every direction of the charge lattice. As shown in [46], this can be usefully formulated as the statement that the convex hull of the charge-to-tension vectors $z_i = q_i/T$ contains the black brane region, as shown in Figure 1.

In general, the size and shape of the black brane region depend on the moduli ϕ^a and their coupling to the gauge fields through $\tau(\phi)$. The boundary of the region consists of extremal black branes, which have vanishing self-force [47], i.e., the long-range force between a pair of identical extremal black branes vanishes. This vanishing self-force involves a balance between the repulsive gauge force and the attractive gravitational and scalar forces. Therefore, the presence of moduli ϕ^a can only increase the size of the black brane region in charge-to-tension space [48]. This means that the black brane region will always contain the ball of radius

$$\|Z\|_{\text{RN}} = \sqrt{\frac{p(d-p-2)}{d-2}}, \quad (2.3)$$

which is the size of the “Reissner-Nordström” black brane region in a theory without

massless scalar fields. The WGC thus implies that the convex hull of the charge-to-tension vectors contains this ball as well.

In a theory with a single $U(1)$, the WGC reduces to the simple expression

$$\frac{|n|e_{p;d}}{T} \geq \|Z\|_{\text{ext}} \geq \sqrt{\frac{p(d-p-2)}{d-2}}, \quad (2.4)$$

where $e_{p;d} = 1/\sqrt{\tau}$ is the gauge coupling of the p -form gauge field, $q = n$ is the charge of the brane, and $\|Z\|_{\text{ext}}$ is the charge-to-tension ratio of an extremal black brane.

2.2 The axion WGC

Periodic scalar fields, also known as axions, can be usefully described as 0-form gauge fields, which couple electrically to objects of worldvolume dimension 0, also known as instantons. The axion WGC thus describes the extension of the WGC described in the previous subsection to the case $p = 0$.

For $p = 0$, the action (2.1) takes the form

$$S = \int \sqrt{-g} d^d x \left(\frac{1}{2\kappa_d^2} \mathcal{R} - \frac{1}{2} g_{ab}(\phi) \partial_\mu \phi^a \partial^\mu \phi^b - \frac{1}{2} f_{ij}(\phi) \partial_\mu \theta^i \partial^\mu \theta^j \right). \quad (2.5)$$

Here, each θ^i has periodicity 2π , i.e., $\theta^i \simeq \theta^i + 2\pi$, and we have denoted the axion kinetic matrix by f_{ij} . For the case of a single axion, $f_{ij} = f^2$ is the square of the axion decay constant f .

The charge of an instanton under the axions θ^i is labeled by a vector $q_i \in \mathbb{Z}^n$. Its norm is given by contraction with the inverse axion kinetic matrix:

$$\|q\|^2 = q_i f^{ij} q_j \equiv \frac{1}{f_q^2}. \quad (2.6)$$

By the analogy with the higher-form case $p \geq 1$, we define charge-to-action vectors

$$z_i = \frac{q_i}{S}, \quad \|z\| = z_i f^{ij} z_j = \frac{1}{f_q^2 S^2}. \quad (2.7)$$

For a theory with a single axion, we have $f_{ij} = f^2$, $q = n$, and therefore

$$\|z\| = \frac{|n|}{fS}. \quad (2.8)$$

We would then like to define the axion WGC as the statement that the charge-to-action vectors z_i of the instantons of the theory must contain the gravitational instanton region. Here, a problem arises: there are three types of gravitational instantons (extremal instantons, cored instantons, and wormholes) [49–51], and it is not *a priori*

clear which of them is the correct one to use [15, 52, 53]. Naively setting $p = 0$ in (2.3) yields a trivial bound, indicating that the axion WGC is more subtle to define than its higher-form cousins. For this reason, studies of the axion WGC in string theory (e.g., [54, 55]) have often focused on understanding how the theory avoids *parametric* violations of the bound.

Recently, the papers [26, 27] argued that in a theory of an axion of decay constant f coupled to quantum gravity, there should exist an instanton of charge n and action S satisfying

$$\|z\| \equiv \frac{|n|}{fS} \geq \|Z\|_{\text{GS}} \equiv \frac{2}{\pi} \sqrt{\frac{d-2}{d-1}}. \quad (2.9)$$

Here, the minimal value $\|Z\|_{\text{GS}}$ is determined by half of the charge-to-action ratio of the Giddings-Strominger wormhole [28], as originally proposed for the axion WGC in [7]. In theories with multiple axions, it generalizes to the statement that the convex hull of the charge-to-action vectors z_i must contain the ball of radius $\|Z\|_{\text{GS}}$ [27]. This bound was derived as a way to avoid a pathology associated with a specific finite *imaginary* distance in axion field space.³ This argument seems to fall somewhat short of a full proof of (2.9), as it leaves open the possibility that some physics other than instantons resolves the pathology associated with imaginary distances.

The result (2.9) was derived in the case without massless moduli. However, it is widely believed that massless moduli arise only in theories with preserved supersymmetry. By weakly breaking supersymmetry, we expect that one can give small masses to the scalar fields without substantially altering the other parameters of the theory, in which case the bound (2.9) applies to theories both with and without massless moduli.

Nonetheless, in theories of p -form gauge fields with $p \geq 1$, the WGC leads to a stronger bound in theories with massless moduli than it does in theories without massless moduli [52]. In this paper, we will show that the same is true for the axion WGC: large classes of instantons in the string landscape and supergravity satisfy a strictly stronger bound than (2.9). To this end, we now present relevant details of supergravity and BPS instantons in four dimensions.

2.3 Supergravity and BPS instantons

In a supersymmetric theory of gravity in $d = 4$ dimensions, the scalar fields $\phi^a = t^i$ (also known as “saxions”) and the axions θ^i pair up into complex scalar fields:

$$T^i = t^i - i \frac{\theta^i}{2\pi}. \quad (2.10)$$

³This is distinct from previous work on upper bounds for finite *real* values of the axion in a holographic context [56, 57], or from familiar phase transitions for finite real values of the theta parameter [58–60].

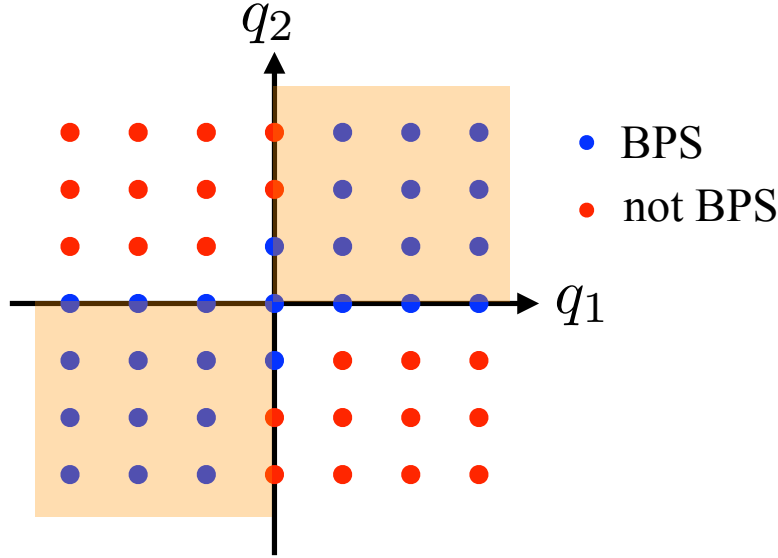


Figure 2: The BPS Cone. BPS instantons charges are constrained to lie inside the BPS cone, which may be identified geometrically with the effective cone of the compactification manifold.

There is thus an equal number of saxions and axions, $n = n_s$, and the saxion kinetic matrix g_{ij} is related to the axion kinetic matrix f_{ij} via

$$g_{ij} = (2\pi)^2 f_{ij} = \frac{1}{2} \partial_i \partial_j \mathcal{K}(t^i), \quad (2.11)$$

where $\mathcal{K}(t^i)$ is the Kähler potential and ∂_i denotes a derivative with respect to the modulus t^i . The shift symmetry of the axions ensures that (within a controlled regime of moduli space), $\mathcal{K}$ depends on θ^i only through non-perturbatively small corrections.

A BPS instanton—also known as a holomorphic instanton in the supersymmetric context—is an instanton whose action is related to its charge q_i via

$$S_q = 2\pi q_i t^i. \quad (2.12)$$

Such instantons are important, among other reasons, because only these instantons can contribute to the superpotential $W(T^i)$, which is a holomorphic function of the complex modulus T^i . Such contributions take the form

$$W(T^i) = \mathcal{A} e^{-2\pi q_i T^i} = \mathcal{A} e^{-S_q + i q_i \theta^i}, \quad (2.13)$$

which notably respects the axion shift symmetry.

In string theory, axions often appear as holonomies of p -form gauge fields over p -cycles of compactification manifolds [61–63]:

$$\theta^i = \int_{\Sigma_p^i} C_p. \quad (2.14)$$

See [64] for a detailed discussion of these extra-dimensional axions. In such compactifications, BPS instantons arise from charged, BPS $(p-1)$ -branes wrapped over calibrated p -cycles. The charges of BPS instantons reside within a particular cone inside the charge lattice known (in different contexts) as the effective cone, the Mori cone, or simply the BPS cone. See Figure 2.

3 Axion WGC from Dualities

In this section, we review the argument of [8], which derived bounds on various axion sectors in string compactifications using higher-form versions of the WGC and the duality web. This approach of using well-understood versions of the WGC to constrain geometry, which in turn can imply less well-understood versions of the WGC, was dubbed the “geometric weak gravity conjecture” in [65].

We begin by considering M-theory on a Calabi-Yau threefold. This theory has electrically charged particles associated with M2-branes wrapped on 2-cycles and magnetically charged strings associated with M5-branes wrapped on 4-cycles.

Theory A			Theory B			$ z _{\min}$
Theory	Branes	WGC for	Theory	Branes	WGC for	
M-theory	M2	5d particles	Type IIB	F1	4d inst.	$\sqrt{\frac{2}{3}}$
M-theory	M2	5d particles	Type IIB	D1	4d inst.	$\sqrt{\frac{2}{3}}$
M-theory	M5	5d strings	Type IIB	D3	4d inst.	$\sqrt{\frac{2}{3}}$
Type IIB	D3	4d particles	Type IIA	D2	4d inst.	$\sqrt{\frac{1}{2}}$

Table 1: Axion WGC via the duality web. After compactifying theory A on a Calabi-Yau manifold, the WGC for particles/strings yields a geometric bound on the sizes of various cycles. Compactifying theory B on this same manifold, this geometric bound translates into a WGC bound on instantons charged under axions; namely, the convex hull of the z -vectors of the charged instantons should contain the ball of radius $\|z\|_{\min}$.

In general, the WGC bound for charged $(p-1)$ -branes in d spacetime dimensions is moduli-dependent. However, since massless scalar fields mediate attractive long-range forces, moduli always increase the charge-to-tension ratio Z_{BH} of an extremal black brane relative to the charge-to-tension ratio Z_{RN} of an extremal Reissner-Nordström black brane. Therefore, moduli always strengthen the WGC bound, so any superextremal brane must satisfy⁴

$$\|z\|^2 \equiv \frac{q^2}{T^2} \geq Z_{\text{BB}}^2 \geq Z_{\text{RN}}^2 = \frac{p(d-p-2)}{d-2}. \quad (3.1)$$

For both particles and strings in 5 dimensions, this yields

$$\|z\| \geq \sqrt{\frac{2}{3}}. \quad (3.2)$$

In an M-theory compactification on a Calabi-Yau threefold, the WGC for particles translates into a bound on the volumes of 2-cycles, while the WGC for strings translates into a bound on the volumes of 4-cycles. Compactifying Type IIB string theory on this same Calabi-Yau, this geometric condition on 2-cycle volumes translates into bounds on worldsheet instantons and D1-brane instantons, and the bound on 4-cycle volumes translates into a bound on D3-brane instantons. In particular, we find that each of these types of instantons must satisfy

$$\|z\| = \frac{1}{f_q S_q} \geq \sqrt{\frac{2}{3}}, \quad (3.3)$$

where the factor of $\sqrt{2/3}$ descends directly from the factor on the right-hand side of (3.2). Note that this argument does not rely on a BPS condition: it applies not only to the holomorphic instantons but to the non-holomorphic ones as well.

A similar argument applies to D2-brane instantons charged under C_3 axions in Type IIA string compactification on Calabi-Yau manifolds. Compactifying Type IIB string theory on the same manifold, one obtains particles from D3-branes wrapping 3-cycles. By (3.1), these particles must satisfy

$$\|z\| \geq \frac{1}{\sqrt{2}}, \quad (3.4)$$

which places constraints on volumes of 3-cycles. Translating these constraints into Type IIA language, we find that D2-brane instantons must satisfy

$$\|z\| = \frac{1}{f_q S_q} \geq \frac{1}{\sqrt{2}}. \quad (3.5)$$

These results are shown in Table 1.

⁴This differs from the value reported in [8] by a factor of $\sqrt{2}$; this discrepancy seems to be due to a difference in conventions in the definition of z .

4 Homogeneous Kähler Potentials

Let us suppose that the Kähler potential takes the general form

$$\mathcal{K} = -\log(\mathcal{F}(t^i)), \quad (4.1)$$

where $\mathcal{F}(t^i)$ is a homogeneous function of degree p , i.e., $\mathcal{F}(\lambda t^i) = \lambda^p \mathcal{F}(t^i)$.

The Kähler metric then takes the form

$$\kappa_{ij} = \partial_i \partial_j \mathcal{K} = \left(\frac{\mathcal{F}_i \mathcal{F}_j}{\mathcal{F}^2} - \frac{\mathcal{F}_{ij}}{\mathcal{F}} \right). \quad (4.2)$$

Here, $\mathcal{F}_i = \partial_i \mathcal{F}$ and $\mathcal{F}_{ij} = \partial_i \partial_j \mathcal{F}$. Using the homogeneity of $\mathcal{F}$, we have

$$t^i \mathcal{F}_i = p \mathcal{F}, \quad t^i \mathcal{F}_{ij} = (p-1) \mathcal{F}_j, \quad \mathcal{F}^{ij} \mathcal{F}_j = \frac{1}{p-1} t^i, \quad (4.3)$$

where $\mathcal{F}^{ij}$ is the inverse of $\mathcal{F}_{ij}$, i.e., $\mathcal{F}^{ij} \mathcal{F}_{jk} = \delta_k^i$.

The inverse metric κ^{ij} is then given by

$$\kappa^{ij} = \left(\frac{1}{p-1} t^i t^j - \mathcal{F} \mathcal{F}^{ij} \right). \quad (4.4)$$

4.1 Positivity lemma

By assumption, the metric κ_{ij} is positive-definite. Under this assumption, we claim that the modified metric

$$h_{ij} = \left(\frac{p-1}{p} \frac{\mathcal{F}_i \mathcal{F}_j}{\mathcal{F}^2} - \frac{\mathcal{F}_{ij}}{\mathcal{F}} \right) \quad (4.5)$$

is positive-semidefinite.

To see this, we first note that we may write

$$\kappa_{ij} = h_{ij} + \xi_{ij}, \quad \xi_{ij} = \frac{1}{p} \frac{\mathcal{F}_i \mathcal{F}_j}{\mathcal{F}^2}. \quad (4.6)$$

We also note that

$$t^i h_{ij} = 0, \quad (4.7)$$

using the identities in (4.3).

From here, the proof proceeds by contradiction. Let us suppose that h_{ij} is *not* positive-semidefinite. Then, there exists v^i such that $v^i h_{ij} v^j < 0$. Without loss of generality, we may decompose v^i as

$$v^i = \alpha t^i + u^i, \quad \text{where } \alpha = \frac{v^i \mathcal{F}_i}{p \mathcal{F}}. \quad (4.8)$$

With this decomposition, we have

$$u^i \mathcal{F}_i = (v^i - \alpha t^i) \mathcal{F}_i = 0. \quad (4.9)$$

We then have

$$0 > v^i h_{ij} v^j = (\alpha t^i + u^i) h_{ij} (\alpha t^j + u^j) = u^i h_{ij} u^j, \quad (4.10)$$

where in the last equality we have used (4.7). From here, we may use the definition of ξ_{ij} and the fact that $u^i \mathcal{F}_i = 0$ to set

$$0 > u^i h_{ij} u^j = u^i (\kappa_{ij} - \xi_{ij}) u^j = u^i \kappa_{ij} u^j. \quad (4.11)$$

This contradicts our assumption that κ_{ij} was positive definite. We conclude that h_{ij} is positive-semidefinite.

4.2 A sharpened WGC

As discussed in §2.3, the metric on axion moduli space is given by

$$f_{ij} = \frac{1}{2(2\pi)^2} \kappa_{ij}. \quad (4.12)$$

A BPS instanton of charge q_i has an action given by

$$S_q = 2\pi q_i t^i. \quad (4.13)$$

The associated charge/decay constant is given by

$$\|q\|^2 \equiv f_q^{-2} = q_i f^{ij} q_j, \quad (4.14)$$

where f^{ij} is the inverse of f_{ij} .

With this, we claim that the axion WGC takes the form

$$f_q S_q \leq \sqrt{\frac{p}{2}}. \quad (4.15)$$

To prove this, we first define

$$\tilde{q}^i = f^{ij} q_j. \quad (4.16)$$

With this,

$$S_q = 2\pi \tilde{q}^i f_{ij} t^j = \frac{1}{4\pi} \frac{\tilde{q}^i \mathcal{F}_i}{\mathcal{F}}, \quad (4.17)$$

while

$$\|q\|^2 = \tilde{q}^i f_{ij} \tilde{q}^j = \frac{1}{2(2\pi)^2} \left(\frac{(\tilde{q}^i \mathcal{F}_i)^2}{\mathcal{F}^2} - \frac{\mathcal{F}_{ij} \tilde{q}^i \tilde{q}^j}{\mathcal{F}} \right). \quad (4.18)$$

We may further rewrite this as

$$\|q\|^2 = \frac{1}{2p(2\pi)^2} \frac{(\tilde{q}^i \mathcal{F}_i)^2}{\mathcal{F}^2} + \frac{1}{2(2\pi)^2} \tilde{q}^i h_{ij} \tilde{q}^j, \quad (4.19)$$

where h_{ij} is the matrix proven to be positive-semidefinite in §4.1. Using this positivity and substituting (4.17), we have

$$\|q\|^2 \geq \frac{1}{2p(2\pi)^2} \frac{(\tilde{q}^i \mathcal{F}_i)^2}{\mathcal{F}^2} = \frac{2}{p} S_q^2. \quad (4.20)$$

Setting $f_q = 1/\|q\|$, we arrive finally at

$$f_q S_q \leq \sqrt{\frac{p}{2}}, \quad (4.21)$$

which completes the proof.

Note that this bound is saturated for an instanton of charge $q_i \propto \mathcal{F}_i$.⁵ To see this, set $q_i = a\mathcal{F}_i$. Then, using the identities (4.3) and (4.4), we have

$$\|q\|^2 = 2 \cdot (2\pi a)^2 \mathcal{F}_i \kappa^{ij} \mathcal{F}_j = 2p \cdot (2\pi a \mathcal{F})^2, \quad S_q = 2\pi q_i t^i = 2\pi a p \mathcal{F}. \quad (4.22)$$

Together, these give

$$f_q S_q = \frac{S_q}{\|q\|} = \sqrt{\frac{p}{2}}, \quad (4.23)$$

so (4.21) is saturated for any BPS instanton of charge $q_i \propto \mathcal{F}_i$, as claimed.

4.3 Generalization to multiple terms

Suppose now that the Kähler potential can be written as a sum of terms of the form in (4.1):

$$\mathcal{K} = - \sum_{\alpha=1}^N \log(\mathcal{F}_\alpha(t_\alpha^i)), \quad (4.24)$$

where $\mathcal{F}_\alpha(\lambda t_\alpha^i) = \lambda^{p_\alpha} \mathcal{F}_\alpha(t_\alpha^i)$ is a homogeneous function of the moduli t_α^i of degree p_α . In this case, the metric assumes a block-diagonal form, and each of the N terms in the prepotential corresponds to a distinct decoupled sector. For an instanton carrying charge q solely in the α th sector, we may disregard all other terms in the sum, yielding

$$(f_q S_q)^2 \leq \frac{p_\alpha}{2}. \quad (4.25)$$

⁵Note that this q_i is a moduli-dependent quantity. However, if we evaluate this charge q_i at a fixed point t_* in moduli space, making it explicitly moduli-independent, we find that it saturates the bound at the point $t = t_*$ in moduli space. It is also not, in general, a rational direction in charge space, but properly quantized instantons can come arbitrarily close to it.

More generally, however, one may consider a multi-sector instanton, i.e., one with charge q whose support lies in multiple sectors. It is not hard to show that, in general, BPS instantons must satisfy

$$f_q S_q \leq \sqrt{\frac{p_{\text{tot}}}{2}}, \quad p_{\text{tot}} = \sum_{\alpha=1}^N p_{\alpha}. \quad (4.26)$$

This bound will be saturated for charges of the form

$$q_i^{\alpha} = a \frac{\partial}{\partial t_{\alpha}^i} [\log (\mathcal{F}_{\alpha})], \quad (4.27)$$

where a is a constant.

4.4 Kähler potentials in the landscape

Kähler potentials that are logarithmic are very common in weakly coupled corners of the string theory landscape. Recently, they have played a central role in a series of papers that discussed EFT strings, that is, axionic strings whose core probes an infinite-distance limit in moduli space [40, 66, 67].

A canonical example of a Kähler potential of the form studied here occurs for C_4 axions in Type IIB O3/O7 orientifold compactifications on Calabi-Yau threefolds. Here, the Kähler potential takes the form

$$\mathcal{K} = -2 \log \mathcal{V}, \quad (4.28)$$

where (at tree level) $\mathcal{V}$ is homogeneous of degree $p = 3/2$ in the 4-cycle volumes τ^i . Thus, $\mathcal{F} = \mathcal{V}^2$ is homogeneous of degree $p = 3$, and

$$f_q S_q \leq \sqrt{\frac{p}{2}} = \sqrt{\frac{3}{2}} \quad (4.29)$$

for C_4 axions in Type IIB O3/O7 orientifold compactifications. Saturation occurs for instantons whose action is proportional to a modulus controlling the overall volume.

Naively, C_2 axions have $p = 6$, since the volume $\mathcal{V}$ is homogeneous of degree 3 in the 2-cycle volumes t^i . However, there is an important subtlety here: in O3/O7 orientifold compactifications, C_2 axions do not reside in the same multiplets as the 2-cycle volume moduli. Said differently, the 2-cycle volumes t^i are not the saxion partners of the C_2 axions. Indeed, the surviving 4-cycle volume moduli τ^i are even under the orientifold action, whereas the surviving C_2 axions are odd.

As a result, the argument used throughout this section breaks down for C_2 axions in O3/O7 orientifolds. (A different argument given in Appendix A of [16] gives $fS \leq 1$

for such odd axions.) However, our argument applies straightforwardly to C_2 axions in O5/O9 orientifolds, as the 2-cycle volumes t^i are the saxion partners of C_2 axions c^i in such orientifolds. As discussed in, e.g., [68], the dilaton and Kähler moduli are mixed in O5/O9 orientifolds relative to O3/O7 orientifolds, and as a result the Kähler potential takes the form

$$\mathcal{K} = -\log \mathcal{V}, \quad (4.30)$$

where (at tree level) $\mathcal{V}$ is cubic in the 2-cycle volumes t^i . Consequently, we have $\mathcal{F} = \mathcal{V}$, $p = 3$, and thus

$$f_q S_q \leq \sqrt{\frac{p}{2}} = \sqrt{\frac{3}{2}}, \quad (4.31)$$

as in the case of C_4 axions above. Saturation occurs for instantons whose action is proportional to the overall volume modulus.

By an extension of the arguments used above, it is straightforward to show directly that in $\mathcal{N} = 2$ compactifications of Type IIB on Calabi-Yau threefolds, the bound $f_q S_q \leq \sqrt{3/2}$ is satisfied by D3-brane instantons, D1-brane instantons, and worldsheet instantons charged respectively under C_4 axions, C_2 axions, and B_2 axions.

Meanwhile, the axio-dilaton $\tau = \exp(-\phi) + i\frac{\theta}{2\pi}$, $\theta = 2\pi C_0$ appears in the Kähler potential via a term of the form

$$\mathcal{K} = -\log(\tau + \tau^\dagger), \quad (4.32)$$

which thus has $p = 1$ and

$$f_q S_q \leq \frac{1}{\sqrt{2}}. \quad (4.33)$$

In fact, this bound is saturated by D(−1)-branes. More generally, 1/2-BPS instantons in maximal supergravity have $f_q S_q = 1/\sqrt{2}$ in all dimensions d [43]. This is closely related to the Sharpened Distance Conjecture [36]. See [69] for further discussion.

In O3/O7 orientifold compactifications, by the argument of §4.3, instantons carrying multi-sector charge under both C_4 axions and the C_0 axion have

$$(f_q S_q)^2 \leq \frac{1}{2} \sum_{\alpha} p_{\alpha} = \frac{1}{2}(3+1) = 2. \quad (4.34)$$

The largest value of p that we have encountered arises in the context of M-theory compactifications on G_2 manifolds. Here, the saxions for C_3 axions appear in a prepotential term with $p = 7$, and hence they satisfy

$$f_q S_q \leq \sqrt{\frac{7}{2}}. \quad (4.35)$$

Theory	Compactification	Axion	p	$\ z\ _{\min}$
Type IIB	O3/O7 Orientifold	C_4	3	$\sqrt{\frac{2}{3}}$
Type IIB	O5/O9 Orientifold	C_2	3	$\sqrt{\frac{2}{3}}$
Type IIB	CY3 Manifold	C_0	1	$\sqrt{2}$
Type IIA	CY3 Manifold	C_3	4	$\sqrt{\frac{1}{2}}$
M-theory	G_2 Manifold	C_3	7	$\sqrt{\frac{2}{7}}$

Table 2: Examples of homogeneous Kähler potentials $\mathcal{K}$. Here, p is the degree of homogeneity of $e^{-\mathcal{K}}$, and $\|z\|_{\min}$ is the smallest charge-to-action ratio of BPS instantons charged under the listed axions in the theory. These values of $\|z\|_{\min}$ agree exactly with those of Table 1 whenever both approaches apply.

The values of p and $\|z\|_{\min}$ for these various examples are shown in Table 2. Comparing with Table 1 above, we see that the values of $\|z\|_{\min}$ match in all cases where both approaches apply (namely, for B_2 , C_2 , and C_4 axions in 4d Type IIB and C_3 axions in 4d Type IIA). This is remarkable because, while the present approach deals exclusively with BPS instantons, the approach used in §3 applies also to directions in the instanton charge lattice outside the BPS cone. Thus, there was *a priori* no reason to expect that the two bounds would agree. We speculate that this agreement may be a general result of 4d supergravity theories in the landscape, and the convex hull of instanton z -vectors *always* contains the ball of radius $\|z\|_{\min} = \sqrt{2/p}$. We leave this as an open question for further study.

4.4.1 Kähler potential splitting

In certain situations, a single term in the Kähler potential may effectively split into a sum of distinct ones, preserving the total $p_{\text{tot}} = \sum_{\alpha} p_{\alpha}$. For instance, while a generic G_2 compactification of M-theory has a single Kähler potential term with $p = 7$, a compactification of M-theory on T^7 (or equivalently, Type II on T^6) yields a sum over 7 terms:

$$\mathcal{K} = - \sum_{\alpha=1}^7 \log(T_{\alpha} + T_{\alpha}^{\dagger}), \quad (4.36)$$

each of which has $p_{\alpha} = 1$, ensuring that the sum $p_{\text{tot}} = \sum_{\alpha} p_{\alpha} = 7$ is preserved. In Type IIA on a Calabi-Yau threefold with O6 planes, the dilaton and complex structure

moduli together appear in a $p = 4$ term while the Kähler moduli appear in a $p = 3$ term [70], again totaling the $p = 7$ degree of the M-theory limit.

Similarly, the Kähler potential term for C_3 axions in Type IIA on a Calabi-Yau threefold has $p = 4$. Under mirror symmetry, this term splits into the aforementioned terms involving C_4 axions and the C_0 axion, which have $p_1 = 3$ and $p_2 = 1$, respectively, ensuring that the sum $\sum_\alpha p_\alpha = 4$ is preserved.

4.5 A bound for axion strings

So far we have used an homogeneous Kähler potential to find the bound (4.15) on fS for instantons. While instantons are electrically charged under axions, axion strings are magnetically charged under axions. In this subsection, we adapt our work so far to put the bound

$$\frac{\mathcal{T}_{\tilde{q}}}{\|\tilde{q}\|} \leq \sqrt{\frac{p}{2}} \quad (4.37)$$

on the tension $\mathcal{T}_{\tilde{q}}$ and magnetic charge $\|\tilde{q}\|$ of an axion string of charge $\tilde{q}$.

From (4.12), the metric on axion moduli space is

$$f_{ij} = \frac{1}{2(2\pi)^2} \kappa_{ij}. \quad (4.38)$$

Above we used this to find that the associated charge/decay constant of an instanton of charge q is given by

$$\|q\|^2 \equiv f_q^{-2} = q_i f^{ij} q_j. \quad (4.39)$$

Similarly, the (magnetic) charge of an axion string of charge $\tilde{q}$ is

$$\|\tilde{q}\|^2 \equiv (2\pi)^2 \tilde{q}^i f_{ij} \tilde{q}^j = \frac{1}{2} \tilde{q}^i \kappa_{ij} \tilde{q}^j. \quad (4.40)$$

The tension, $\mathcal{T}_{\tilde{q}}$, of a BPS axion string of magnetic charge $\tilde{q}^i$ has tension given by [40, 66]

$$\mathcal{T}_{\tilde{q}} = -\frac{1}{2} \tilde{q}^i \partial_i \mathcal{K} = \frac{1}{2} \tilde{q}^i \frac{\mathcal{F}_i}{\mathcal{F}}. \quad (4.41)$$

From §4.1, we know that h_{ij} is positive semi-definite. Using $h_{ij} = \kappa_{ij} - \xi_{ij}$ and contracting with $\tilde{q}^i$ on both sides, we find

$$\tilde{q}^i \kappa_{ij} \tilde{q}^j - \frac{1}{p} \left(\frac{\tilde{q}^i \mathcal{F}_i}{\mathcal{F}} \right)^2 \geq 0. \quad (4.42)$$

Using (4.40), (4.41) and the definition of ξ_{ij} , rearranging and taking the square root, we find (4.37):

$$\frac{1}{\|z^{\text{ax str}}\|} \equiv \frac{\mathcal{T}_{\tilde{q}}}{\|\tilde{q}\|} \leq \sqrt{\frac{p}{2}}. \quad (4.43)$$

Note that this precisely matches the bound (4.21) on $\|z^{\text{inst}}\| = \frac{1}{f_q S_q}$. We conclude that for such homogeneous Kähler potentials, $\|z^{\text{ax str}}\|_{\text{min}} = \|z^{\text{inst}}\|_{\text{min}} = \sqrt{2/p}$.

4.6 Bounds on the Kähler potential

In this subsection, we derive bounds on the degree of homogeneity p —and thus the minimal value $\|z_{\text{min}}\|$ —using two different approaches: (i) universal features of infinite-distance limits and (ii) bounds on imaginary traversals, as initiated in the recent works [26, 27].

4.6.1 Infinite-distance bounds

So far, we have bounded the instanton charge-to-action ratio $\|z\| = 1/(f_q S_q)$ in terms of the degree of homogeneity p . We now wish to bound this degree of homogeneity using asymptotic properties of the moduli space.

We focus our attention on the homogeneous scaling limit $t^i \rightarrow \lambda t^i$, $\lambda \rightarrow \infty$. In this limit, the kinetic term for the scaling parameter λ takes the form

$$-\frac{1}{4}\kappa_{\lambda\lambda}\partial_\mu\lambda\partial^\mu\lambda = -\frac{1}{4}\frac{\partial^2\mathcal{K}}{\partial\lambda^2}\partial_\mu\lambda\partial^\mu\lambda = -\frac{p}{4\lambda^2}\partial_\mu\lambda\partial^\mu\lambda. \quad (4.44)$$

By supersymmetry, the decay constant f_λ of the axion partner of the scalar field λ takes the form

$$f_\lambda^2 = \frac{1}{2(2\pi)^2}\kappa_{\lambda\lambda} \sim \frac{1}{\lambda^2}. \quad (4.45)$$

The Weak Gravity Conjecture for strings [1] implies that in the limit $\lambda \rightarrow \infty$, there must exist an axion string with tension bounded above as

$$\mathcal{T} \lesssim f_\lambda \sim \frac{1}{\lambda}. \quad (4.46)$$

Written in terms of the canonically normalized field $\hat{\lambda} = \sqrt{p/2}\log\lambda$, we further have

$$\mathcal{T} \lesssim \frac{1}{\lambda} = \exp\left(-\sqrt{\frac{2}{p}}\hat{\lambda}\right). \quad (4.47)$$

Next, we require the axion string scale $M_{\text{as}} \equiv \sqrt{2\pi\mathcal{T}}$ to lie above the species scale Λ_{QG} , which is expected to be true of all EFT strings [71], i.e., those associated with infinite-distance limits [66]. According to various swampland conjectures [36, 72–75], this scale decays exponentially with proper field distance as

$$\Lambda_{\text{QG}} \sim \exp(-\alpha_{\text{QG}}\hat{\lambda}), \quad \frac{1}{\sqrt{(d-1)(d-2)}} \leq \alpha_{\text{QG}} \leq \frac{1}{\sqrt{d-2}}, \quad (4.48)$$

where here $d = 4$ is the number of spacetime dimensions.

To ensure $M_{\text{as}} \gtrsim \Lambda_{\text{QG}}$, we must have

$$\sqrt{\frac{2}{p}} \leq \alpha_{\text{QG}}. \quad (4.49)$$

Together with (4.48), this implies a lower bound on the degree of homogeneity p :

$$p \geq 1. \quad (4.50)$$

Placing an upper bound on the degree of homogeneity p is somewhat more difficult. In general, barring the existence of a decoupling rigid field theory [76, 77], we expect that the string WGC bound will be approximately saturated in an infinite-distance limit:

$$\mathcal{T} \sim f_\lambda. \quad (4.51)$$

In special cases, the axion string scale $M_{\text{as}} = \sqrt{2\pi\mathcal{T}}$ may further be parametrically identified with the species scale:

$$M_{\text{as}} \sim \Lambda_{\text{QG}} \sim \sqrt{f_\lambda} \sim \frac{1}{\sqrt{\lambda}} \sim \exp\left(-\sqrt{\frac{1}{2p}}\hat{\lambda}\right). \quad (4.52)$$

In these special cases, (4.48) further implies an upper bound:

$$p \leq 3. \quad (4.53)$$

In the following examples, we will study several examples of Kähler potentials that arise in known corners of the landscape. We indeed find $p \geq 1$ in all examples in the landscape with which we are familiar. Many axion sectors satisfy (and saturate) the upper bound $p \leq 3$, but others violate it.

A more general upper bound $p_{\text{tot}} \leq 7$ can be obtained in the case that the limit $\lambda \rightarrow \infty$ represents a decompactification limit to 10d string theory or 11d M-theory. The Emergent String Conjecture [72] suggests that such a case can always be realized. We will return to this in §5.2 after developing the relevant background material.

In all of the examples considered above, p is an integer. That p is an integer between 1 and 7 in string theory examples was previously emphasized in [40, 66, 67, 71]. In [66], the integrality of p was linked to a conjectured integral scaling relation between membrane tensions and string tensions. Using taxonomy rules for infinite-distance limits developed in [44, 45], one can argue that this is true more generally, as will be shown in forthcoming work [78]. Interestingly, coefficients of the form $\sqrt{2/p}$ for integer p also appear in many examples in the exponential relationship between canonically normalized moduli and other quantities, such as the mass scale of the lightest tower or the potential energy; see, e.g., [33, 66, 79–86].

4.6.2 A bound from imaginary traversals

Suppose that we have some field ϕ with a kinetic term proportional to $-\lambda^\gamma |d\phi|^2$ for a scalar λ with a kinetic term of the form $-\frac{p}{4\lambda^2 \kappa_d^2} \partial_\mu \lambda \partial^\mu \lambda$, and γ a non-zero real constant. Consider the analytic continuation $\lambda = \lambda_0 \exp(i\alpha)$. The theory will break down when $\text{Re } \lambda^\gamma < 0$, i.e., when $|\alpha| > \pi/(2|\gamma|)$. The distance in field space associated with changing α between the extremes $-\pi/(2|\gamma|)$ and $+\pi/(2|\gamma|)$ is $\sqrt{p/2} \Delta\alpha/\kappa_d = \sqrt{p/2} \pi/(\kappa_d |\gamma|)$. If we demand that theory breaks down due to a wrong-sign kinetic term before the imaginary distance bound [26, 27] is reached, we have:

$$\sqrt{\frac{p}{2}} \frac{\pi}{|\gamma| \kappa_d} \leq \pi \sqrt{\frac{d-1}{d-2}} \frac{1}{\kappa_d} \Rightarrow p \leq \frac{2(d-1)}{d-2} \gamma^2. \quad (4.54)$$

This is somewhat speculative: the imaginary distance bound requires that some physics should invalidate the theory before a particular imaginary value of the field displacement is reached. It is not obvious that the only way for this to happen is for the kinetic term of some other field to have the wrong sign. Nonetheless, it is suggestive that we can obtain reasonable bounds, consistent with examples, from this approach.

Let us fix $d = 4$ and consider two cases of interest. First, suppose that the field ϕ is an axion θ that pairs up with λ in a chiral supermultiplet with scalar component $\lambda + i\theta$. Then the kinetic term for θ has $\gamma = -2$ (by supersymmetry), and the bound becomes

$$p \leq 12. \quad (4.55)$$

This is consistent with our examples. A more interesting bound comes from considering ϕ to be a gauge field A with a holomorphic coupling to a scalar field $\lambda + i\theta$. Here the power of λ is fixed to be $\gamma = 1$ by the periodicity of θ .⁶ In this case, we obtain

$$p \leq 3. \quad (4.56)$$

This is nontrivially realized in many examples, as shown in Table 2 above. In M-theory on G_2 manifolds, we have $p = 7$, but the modulus λ controls a 3-cycle volume, and there is no 6-brane we can wrap on a 3-cycle to obtain 4d fields with kinetic term given directly by λ . Thus, it is unsurprising that (4.56) is violated in these compactifications.

A more general bound on p_{tot} can be obtained directly from the result (1.2) [26, 27], which for $d = 4$ states that

$$f_q S_q \leq \frac{\pi}{2} \sqrt{\frac{3}{2}} \approx 1.92 M_{\text{Pl}}. \quad (4.57)$$

⁶There is an exception in special cases where $\theta \mapsto \theta + 2\pi$ is accompanied by a nontrivial monodromy on the gauge fields themselves. This occurs for Kaluza-Klein gauge theory (see, e.g., [12]), which realizes $\gamma = 3$, again consistent with the bound.

Setting $p_{\text{tot}} = 7$, our bound (4.21) is saturated when

$$f_q S_q = \sqrt{\frac{7}{2}} \approx 1.87 M_{\text{Pl}}, \quad (4.58)$$

which is consistent with (4.57).

In contrast, an example with $p_{\text{tot}} \geq 8$ would have BPS instantons with $f_q S_q \geq 2$, in violation of (4.57). It is a logical possibility that there are non-BPS instantons that have smaller action than BPS instantons and satisfy (4.57), but in the cases that we are aware of where non-BPS instantons have smaller action, they are singular solutions that cannot be uplifted to non-singular solutions in the UV completion [50, 51]. If we assume that non-BPS instantons of lower action than BPS instantons do not arise in physically consistent settings, this again provides an argument for the upper bound $p_{\text{tot}} \leq 7$.

4.6.3 Summary of bounds on p

In summary, then, swampland constraints on infinite-distance limits lead us to expect that the degrees of homogeneity p_α , and hence also their sum p_{tot} , of a Kähler potential of the form (4.24) to be positive integers. When there is only one term in the sum, as in Kähler potentials of the form (4.1), this reduces to the condition that p be a positive integer.

Placing an upper bound is more difficult, but the smallest universal upper bound on p_{tot} , and hence each individual p_α , consistent with known examples is 7 (inclusive). We shall give a partial argument based on infinite distance for this upper bound of 7 in §5.2, once we have developed the necessary background material in the next section.

5 A Bound on α -vectors

In a supersymmetric theory with moduli t^i , we may define the α -vector of an instanton with action S via

$$\alpha_i = \partial_i \log S. \quad (5.1)$$

The length of such an α -vector is given by contraction with the moduli space metric g^{ij} :

$$\|\alpha\|^2 = \alpha_i g^{ij} \alpha_j. \quad (5.2)$$

Recall from §2.3 that in a supersymmetric theory, the axion and saxion form the real and complex parts of a scalar field:

$$T = t - i \frac{\theta}{2\pi}. \quad (5.3)$$

This gives a relation between the kinetic terms for θ and t :

$$g_{tt} = (2\pi)^2 f_{\theta\theta} = (2\pi f)^2. \quad (5.4)$$

A BPS instanton of charge $q = n$ has an action given by

$$S = 2\pi |n| t, \quad (5.5)$$

so

$$\|\alpha\| = \sqrt{g^{tt}} |\partial_t \log S| = \sqrt{g^{tt}} \frac{2\pi |n|}{S} = \sqrt{f_{\theta\theta}} \frac{|n|}{S} = \frac{|n|}{fS}. \quad (5.6)$$

And thus,

$$\frac{|n|}{fS} = \|\alpha\|. \quad (5.7)$$

This generalizes straightforwardly to supersymmetric theories with multiple axions. Set $T^i = t^i - i \frac{\theta^i}{2\pi}$, and consider a supersymmetric instanton of charge vector q_i . This instanton will have action

$$S_q = 2\pi |q_i t^i|, \quad (5.8)$$

and it will feel an effective decay constant of

$$f_q^2 \equiv \frac{1}{\|q\|^2} = \frac{1}{q_i f^{ij} q_j}. \quad (5.9)$$

As discussed in §2.3, the metric on axion moduli space f_{ij} is related to the metric on saxion moduli space g_{ij} via

$$f_{ij} = \frac{1}{(2\pi)^2} g_{ij}. \quad (5.10)$$

We thus have

$$\|\alpha\|^2 = g^{ij} \partial_i (\log S) \partial_j (\log S) = \frac{1}{S_q^2} g^{ij} (2\pi q_i) (2\pi q_j) = \frac{1}{S_q^2} f^{ij} q_i q_j = \frac{1}{f_q^2 S_q^2} = \|z\|^2. \quad (5.11)$$

In fact, there are explicit BPS “EFT instanton” solutions that realize all of these properties, which are conveniently written in terms of the dual saxion fields [66, 71].

The upshot of this is that a bound on the magnitude of the α -vector $\|\alpha\|$ of a BPS instanton immediately translates into a bound on the magnitude of the charge-to-action vector $\|z\|$. In this section, we derive a bound on the former.

5.1 Bound on $\|\alpha\|$

In this subsection, we find a bound on $\|\alpha\|$ for BPS instantons in various string compactifications. The key observation enabling this inference is the scaling behavior of wrapped brane tensions on the volume modulus, ρ . In general, a $(p-1)$ -brane in d dimensions that descends from a wrapped $(P-1)$ -brane in D dimensions has a tension that scales with the canonically normalized volume modulus ρ as [45]

$$T_p \sim \exp \left(-\frac{p(D-2) - P(d-2)}{\sqrt{(D-d)(D-2)(d-2)}} \rho \right). \quad (5.12)$$

For an instanton, we have $p = 0$, and the “tension” T_0 is the instanton action

$$S \sim \exp \left(\frac{P(d-2)}{\sqrt{(D-d)(D-2)(d-2)}} \rho \right). \quad (5.13)$$

In addition, a $D(p-1)$ -brane in Type II string theory has a tension that scales with the canonically normalized 10d dilaton ϕ as

$$T_p \sim \exp \left(\frac{4-p}{\sqrt{8}} \phi \right), \quad (5.14)$$

while the fundamental string tension scales as

$$T_2 \sim \exp \left(-\frac{1}{\sqrt{2}} \phi \right). \quad (5.15)$$

As a result, the volume modulus ρ and the dilaton ϕ contribute universally to $\vec{\alpha}$, with

$$\alpha_\rho = \partial_\rho \log S, \quad \alpha_\phi = \partial_\phi \log S. \quad (5.16)$$

These give a lower bound on the magnitude of the α -vector:

$$\|\alpha\|^2 \geq \alpha_\rho^2 + \alpha_\phi^2. \quad (5.17)$$

This bound will be saturated if the instanton action depends only on the dilaton and volume modulus. In general, however, we expect additional contributions to $\|\vec{\alpha}\|$ from the “shape moduli” of the compactification manifold.

As a first example, let us consider the C_4 axions ϑ_i of Type IIB string theory compactified on a Calabi-Yau threefold, which are associated with the holonomy of C_4 over the 4-cycles $\Sigma_4^{(i)}$. Charged instantons come from D3-branes wrapped over the 4-cycles. By (5.13) actions of these instantons scale with the volume modulus as

$$S \sim \exp \left(\sqrt{\frac{2}{3}} \rho \right) \quad \Rightarrow \quad \alpha_\rho = \sqrt{\frac{2}{3}}. \quad (5.18)$$

Theory	Dimension	Axions	α_ϕ	α_ρ	$\ \alpha_{\min}\ $
Type IIB	4	C_0	$\sqrt{2}$	0	$\sqrt{2}$
		B_2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{6}}$	$\sqrt{\frac{2}{3}}$
		C_2	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{6}}$	$\sqrt{\frac{2}{3}}$
		C_4	0	$\sqrt{\frac{2}{3}}$	$\sqrt{\frac{2}{3}}$
	6	C_0	$\sqrt{2}$	0	$\sqrt{2}$
		B_2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
		C_2	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
Type IIA	4	B_2	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{6}}$	$\sqrt{\frac{2}{3}}$
		C_3	$\sqrt{\frac{1}{8}}$	$\sqrt{\frac{3}{8}}$	$\frac{1}{\sqrt{2}}$
M-theory	4	C_3	—	$\sqrt{\frac{2}{7}}$	$\sqrt{\frac{2}{7}}$

Table 3: Contributions to $\|\alpha\|$ from the dilaton and radion for instantons (and axion strings) charged electrically (magnetically) under various types of axions in string/M-theory compactifications. In all axion sectors where the approaches from the previous sections apply, the value of $\|\alpha\|_{\min}$ obtained here exactly matches the values of $\|z\|_{\min}$ obtained there, as shown in Tables 1 and 2.

The D3-brane tension is independent of the 10d dilaton ϕ , so $\alpha_\phi = 0$. Thus, by (5.17), we see that D3-brane instantons have

$$\|\alpha\| \geq |\alpha_\rho| = \sqrt{\frac{2}{3}}. \quad (5.19)$$

This precisely matches the minimal value $\|z\|_{\min}$ for ED3-brane instantons charged under C_4 axions obtained in §3 and §4 above.

Next, let us consider the case of the B_2/C_2 axions of Type IIB string theory on a Calabi-Yau threefold, which come from the holonomies of these 2-form fields over 2-cycles. Charged instantons come from wrapped fundamental strings and wrapped D1-branes, respectively. By (5.13), these both have

$$S \sim \exp\left(\frac{1}{\sqrt{6}}\rho\right) \quad \Rightarrow \quad \alpha_\rho = \frac{1}{\sqrt{6}}. \quad (5.20)$$

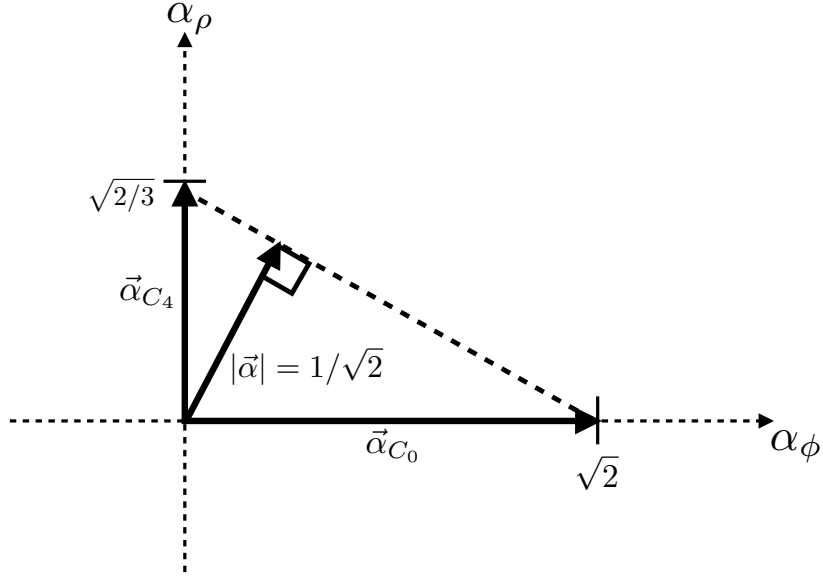


Figure 3: α -vectors for instantons carrying charge under the C_0 axion and C_4 axions. The minimal length α -vector corresponds to the length of the perpendicular bisector, $\|\vec{\alpha}\| = 1/\sqrt{2}$. This matches the minimal value allowed for D2-brane instantons in Type IIA compactifications on Calabi-Yau manifolds, as expected under mirror symmetry.

In addition, by (5.14) and (5.15), the F-string and D-string instantons respectively have

$$S \sim \exp\left(\mp \frac{1}{\sqrt{2}}\phi\right) \quad \Rightarrow \quad \|\alpha_\phi\| = \frac{1}{\sqrt{2}}. \quad (5.21)$$

By (5.17), we therefore have

$$\|\alpha\|^2 \geq \alpha_\rho^2 + \alpha_\phi^2 = \frac{1}{6} + \frac{1}{2} = \frac{2}{3}. \quad (5.22)$$

Once again, this precisely matches the value of $\|z\|_{\min}$ obtained through the approaches in §3 and §4 above.

This procedure may be repeated for the other types of axions/instantons in Type II compactifications on Calabi-Yau threefolds and K3 surfaces. The results are shown in Table 3. Comparing with Tables 1 and 2, we see that the three approaches to bounding $\|z\|$ agree whenever two or more apply.

For C_3 axions in Type IIA string theory on a Calabi-Yau threefold, $\|z\| = \|\alpha\|$ is bounded below by $\|\alpha_{\min}\| = 1/\sqrt{2}$. This can be understood via mirror symmetry as follows: under mirror symmetry, both C_4 axions and the universal C_0 axion of Type IIB are related to C_3 axions of Type IIA. From Table 3, we see that C_4 and C_0 axions

have respective α -vectors of

$$\vec{\alpha}_{C_4} = \left(0, \sqrt{\frac{2}{3}}\right), \quad \vec{\alpha}_{C_0} = (\sqrt{2}, 0). \quad (5.23)$$

Instantons carrying multi-sector charge under both C_0 and C_4 axions will have α -vectors that reside on the line segment between these two vectors. As shown in Figure 3, the minimum length of an α -vector along this line segment is $\|\alpha\| = 1/\sqrt{2}$, which matches the minimal value for C_3 axions in Type IIA. Said differently, the minimum value of $\|z\| = \|\alpha\| = 1/\sqrt{2}$ for D2-brane instantons in Type IIA compactifications corresponds, under mirror symmetry, to a diagonal direction in the instanton lattice generated by D3-brane instantons and D(-1)-brane instantons. This is the α -vector manifestation of the prepotential splitting discussed in §4.4.1.

Finally, note that the same procedure yields $\|\alpha_{\min}\| = \sqrt{2/7}$ for instantons associated with M2-branes wrapped on 3-cycles of G_2 manifolds, in agreement with the bound from homogeneous Kähler potentials in the previous section. This is the smallest value of $\|z\| = \|\vec{\alpha}\|$ that we have observed in the landscape. In the following subsection, we will argue that $\|z\| = \sqrt{2/7}$ also represents the smallest value of $\|z\|$ that can be obtained for BPS instantons with homogeneous Kähler potentials.

5.2 Toward an upper bound on p

In §4.6.1, we argued that the degree of homogeneity p of a homogeneous Kähler potential is bounded below by $p \geq 1$. In what follows, we will argue that under certain assumptions, p is bounded above by $p \leq 7$. By the argument of §4, this in turn implies an upper bound $f_q S_q \leq \sqrt{7/2}$ on BPS instantons in such theories.

We begin with the case of a single homogeneous sector, so that the Kähler potential takes the form (4.1):

$$\mathcal{K} = -\log(\mathcal{F}(t^i)). \quad (5.24)$$

We fix a starting point $t = t_*$, and consider the path in moduli space $t = \lambda t_*$, $\lambda \rightarrow \infty$. According to the Emergent String Conjecture, when this limit is sensible, it represents either an emergent string limit or a decompactification limit. The classical infinite distance limit may be obstructed by quantum effects [87–89], in which case we expect that we can consider homogeneous scaling of additional sectors of moduli to obtain a valid infinite-distance limit. The argument below generalizes straightforwardly to such a setting.

We now consider the charge

$$q_i = \frac{a\lambda\mathcal{F}_i}{p\mathcal{F}} \bigg|_{t=\lambda t_*}, \quad (5.25)$$

for constant a , and assume that there is a BPS instanton of this charge.⁷ Note that due to us evaluating the expression at $t = \lambda t_*$, and the factors of λ canceling, this q is independent of λ , so is indeed a moduli-independent quantity (in other words, it is computed in terms of a choice of t_* , but is then held fixed as we consider points other than t_* in moduli space.) But, due to its definition, everywhere along the path $t = \lambda t_*$, $q_i \propto \mathcal{F}_i/\mathcal{F}$. We proved in §4.2 that such a charge saturates the bound (4.15). Using this, and that $\|\alpha\| = \|z\|$, we find that along the path $t = \lambda t_*$, this charge satisfies

$$\sqrt{\frac{2}{p}} = \|z\| = \|\alpha\|. \quad (5.26)$$

Hence if we can bound $\|\alpha\| \geq \alpha_{\min}$, then we can bound $p \leq \frac{2}{\alpha_{\min}^2}$.

If the infinite distance limit $\lambda \rightarrow \infty$ represents a decompactification limit to 10d or 11d, then in some duality frame we are dealing with a weakly-coupled compactification of string theory or M-theory. The BPS instantons in 4d must descend from BPS $(k-1)$ -branes charged under k -form gauge fields. These branes are simply the familiar branes from string/M-theory, which were studied in §5.1 and are listed in Table 3. The smallest value of $\|\alpha\|$ we encountered there is $\sqrt{2/7}$, corresponding to M2-branes wrapped on 3-cycles of a G_2 manifold. Thus, if $\lambda \rightarrow \infty$ represents a decompactification limit to 10d or 11d, we must have $p \leq 7$.

If $\lambda \rightarrow \infty$ instead represents a decompactification limit to $D < 10$ spacetime dimensions, we expect that there is a *different* infinite distance limit that we could have taken, scaling additional moduli to enlarge additional dimensions to reach $D = 10$ or $D = 11$. Even with that assumption, our argument has a loophole when applied to p_{tot} . There may be additional moduli that were spectators to the infinite distance limit that we took, which contribute to p_{tot} but not to the value of p that we computed. In examples that we know, this does not lead to exceptions to the bound: for instance, in Type IIB O3/O7 orientifold compactifications, the decompactification limit to 10d involves the Kähler moduli T_i which have $p = 3$, while the complex structure moduli and axiodilaton that are spectators to this limit have $p = 4$, for $p_{\text{tot}} = 7$. Our argument did not constrain the complex structure moduli, but the example still satisfies the bound.

⁷This is not a very good assumption, since the charge q is not in general properly quantized. However, one could run the same argument with the charge $q_i = a \left[\frac{\lambda \mathcal{F}_i}{p \mathcal{F}} \Big|_{t=\lambda t_*} + \eta_i(a) \right]$, where $\eta_i(a)$, a function of a , is the smallest number that results in q_i being integrally quantized. Assuming that there is a BPS instanton of this charge is now more reasonable, since it is properly quantized. Running the argument with this charge, we find that factors of λ cancel, so that this charge has $\|z\| = \sqrt{\frac{2}{p}}(1 + \mathcal{O}(\epsilon))$ independent of λ , for $\epsilon(a) := \max_i |\eta_i|$. ϵ can be made arbitrarily small by making a larger, so we find that the final result, that $p \leq \frac{2}{\alpha_{\min}^2}$, still holds.

The expectation that $p_{\text{tot}} \leq 7$ is closely related to the integral scaling conjecture of [66], as will be discussed in the forthcoming work [78].

5.3 Axion strings

The relation between $\|\alpha\| = \|z\|$ extends to BPS axion strings as well. For an axion string of tension $\mathcal{T}$, we define⁸

$$\alpha_i = -\partial_i \log \mathcal{T}, \quad \|\alpha\|^2 = \alpha_i g^{ij} \alpha_j. \quad (5.27)$$

From (4.40) and (5.10), we have

$$\|z\|^2 = \frac{\|\tilde{q}\|^2}{\mathcal{T}^2} = (2\pi)^2 \frac{\tilde{q}^i f_{ij} \tilde{q}^j}{\mathcal{T}^2} = \frac{\tilde{q}^i g_{ij} \tilde{q}^j}{\mathcal{T}^2}. \quad (5.28)$$

A BPS axion string satisfies

$$\mathcal{T} = -\frac{1}{2} \tilde{q}^i \partial_i \mathcal{K}, \quad (5.29)$$

hence

$$\partial_i \mathcal{T} = -\frac{1}{2} \tilde{q}^j \partial_i \partial_j \mathcal{K} = -\frac{1}{2} \tilde{q}^j \kappa_{ij} = -\tilde{q}^j g_{ij}. \quad (5.30)$$

So

$$\|\alpha\|^2 = \frac{\tilde{q}^i g_{ij} g^{jk} g_{kl} \tilde{q}^l}{\mathcal{T}^2} = \frac{\tilde{q}^i g_{ij} \tilde{q}^j}{\mathcal{T}^2}. \quad (5.31)$$

Hence,

$$\|\alpha\|^2 = \|z\|^2. \quad (5.32)$$

In Type IIB string theory in 4d, axion strings charged magnetically under C_0 , B_2 , C_2 , and C_4 axions come from wrapped D7-branes, NS5-branes, D5-branes, and D3-branes, respectively. For Type IIA string theory in 4d, axion strings charged magnetically under C_3 axions come from D4-branes wrapped on 3-cycles. For M-theory on a G_2 manifold, axion strings charged magnetically under C_3 axions come from M5-branes wrapped on 4-cycles. With the sign convention used in (5.27), the values of α_ρ and α_ϕ obtained for each of these magnetically charged axion strings precisely match the values of α_ρ and α_ϕ obtained for the corresponding electrically charged instantons. Consequently, the value of $\|\alpha\|_{\text{min}} = \sqrt{\alpha_\rho^2 + \alpha_\phi^2}$ for the axion strings precisely matches that of the corresponding instantons, so Table 3 applies equally well to both instantons and axion strings.

⁸Note that we adopt opposite sign conventions for the α vector of an instanton, $\alpha_i = +\partial_i S$, compared to an axion string, $\alpha_i = -\partial_i \mathcal{T}$. This is so that the sign of the quoted values of α_ρ and α_ϕ in Table 3 are correct for both instantons and axion strings.

Putting these results together, we have

$$\|z^{\text{ax str}}\|_{\min} = \|\alpha^{\text{ax str}}\|_{\min} = \|\alpha^{\text{inst}}\|_{\min} = \|z^{\text{inst}}\|_{\min}, \quad (5.33)$$

which fits perfectly with the result

$$\|z^{\text{ax str}}\|_{\min} = \|z^{\text{inst}}\|_{\min} = \sqrt{\frac{2}{p}} \quad (5.34)$$

obtained for homogeneous Kähler potentials in §4.2 and §4.5.

6 Axion Inflation

In this section, we comment briefly on the implications of our work for axion inflation.

Natural inflation was first proposed in [90] as a candidate mechanism for generating super-Planckian inflaton field ranges. The key idea is that the discrete axion shift symmetry protects the axion potential from problematic Planck-suppressed operators, which would destroy the flatness of the potential.

However, super-Planckian natural inflation requires super-Planckian decay constants, which are difficult to construct in string theory [91]. This can be understood from the Weak Gravity Conjecture, as observed in the original paper [1]. For a natural inflation potential of the form

$$V(\theta) = \Lambda_{\text{UV}}^4 e^{-S} (1 - \cos \theta) + O(e^{-2S}), \quad (6.1)$$

natural inflation requires $f \gtrsim 5M_{\text{Pl}}$, while control of the instanton expansion typically requires $S \gg 1$. As we have seen, the axion WGC requires $fS \lesssim M_{\text{Pl}}$, which is in tension with this bound.

In response to these constraints, several ideas were proposed for generating an effective super-Planckian decay constant in string theory. The first, known as N -flation [2, 3], involves recruiting N axions with order-one decay constants, then traversing the N -dimensional space diagonal in axion moduli space to generate an effective decay constant of $O(\sqrt{N})$. The second, known as alignment [4], uses mixing between two or more axions to generate a large effective decay constant. This can either take the form of kinetic mixing or of mass mixing; here, we will focus on the former.

In general, these multi-axion models also run into trouble with the WGC, as discussed in [5–9]. Surveys of the string landscape have also failed to find successful examples of N -flation or axion alignment (see, e.g., [92–97]). In this section, we revisit N -flation and kinetic alignment in light of the precise bounds derived in the present work. In short, we will argue that simple, isotropic models of N -flation are ruled out by our bounds, while some models of kinetic alignment survive.

6.1 Bounds on N -flation

Consider a theory with N axions, and take the axion kinetic matrix to be given simply by

$$\kappa_{ij} = f^2 \delta_{ij}. \quad (6.2)$$

Then, the effective decay constant is given by

$$f_{\text{eff}} = \sqrt{N} f, \quad (6.3)$$

which is the length of the space diagonal of an N -dimensional hypercube of side length f .

Consider the instanton of charge $q_i = (1, 1, \dots, 1)$. This instanton has action $S_q = q_i S^i = \sum_i S^i > N$, since perturbative control requires $S^i > 1$ for all i . The charge of the instanton is given by

$$\|q\| = \sqrt{q_i \kappa^{ij} q_j} = \frac{\sqrt{N}}{f}. \quad (6.4)$$

The BPS axion WGC derived in (4.21) then gives

$$\frac{1}{\sqrt{N} f} \geq \frac{\|q\|}{S_q} \geq \sqrt{\frac{2}{p}}. \quad (6.5)$$

This yields

$$f_{\text{eff}} \leq \sqrt{\frac{p}{2}}, \quad (6.6)$$

which in particular is independent of N . Using multiple axion sectors, the degree of homogeneity p can be replaced by the sum p_{tot} , as in (4.21). However, we have argued above that $p_{\text{tot}} \leq 7$, which precludes parametrically super-Planckian decay constants.

6.2 Bounds on Kinetic Alignment

Axion alignment is much more subtle, and in general, it is not ruled out by the BPS axion WGC. However, certain types of alignment are constrained, as we explain below.

Consider a theory with two axions with axion kinetic matrix

$$\kappa_{ij} = f^2 \begin{pmatrix} 1 & 1 - \epsilon \\ 1 - \epsilon & 1 \end{pmatrix}. \quad (6.7)$$

Kinetic alignment occurs in the limit $\epsilon \rightarrow 0$.

Let us suppose that the BPS cone is generated by two instantons of respective charges $q_i^{(1)} = (1, 0)$, $q_i^{(2)} = (0, 1)$. The diameter $\mathcal{D} = 2\pi f_{\text{eff}}$ of the axion fundamental

domain is given by the length of the diagonal between $\theta_1 = \theta_2 = -\pi$ and $\theta_1 = \theta_2 = +\pi$, which is equal to

$$\mathcal{D} = 2\pi f_{\text{eff}} = \int_{-\pi}^{\pi} dt \sqrt{\kappa_{ij} \dot{\theta}^i \dot{\theta}^j} = 4\pi f \sqrt{1 - \frac{\epsilon}{2}}. \quad (6.8)$$

Let us now suppose that the instanton of charge $q_i = (1, 1)$ has action given by $S_{1,1} = S_1 + S_2 \geq 2$ and satisfies the axion WGC bound (4.21). We then have

$$\sqrt{\frac{2}{p}} \leq \|z\| \leq \frac{\sqrt{q_i \kappa^{ij} q_j}}{2} = \frac{1}{2f \sqrt{1 - \frac{\epsilon}{2}}} = \frac{1}{f_{\text{eff}}}. \quad (6.9)$$

We see that in this scenario, the effective decay constant is bounded above as $f_{\text{eff}} \leq \sqrt{p/2}$, so alignment fails to produce a super-Planckian traversal.

However, the story is different if we instead consider a kinetic matrix of the form

$$\kappa_{ij} = f^2 \begin{pmatrix} 1 & \epsilon - 1 \\ \epsilon - 1 & 1 \end{pmatrix}. \quad (6.10)$$

Once again, alignment occurs in the limit $\epsilon \rightarrow 0$.

Let us again suppose that the BPS cone is generated by two instantons of respective charges $q_i^{(1)} = (1, 0)$, $q_i^{(2)} = (0, 1)$. The diameter $\mathcal{D} = 2\pi f_{\text{eff}}$ of the axion fundamental domain is now given by the length of the diagonal between $\theta_1 = -\theta_2 = -\pi$ and $\theta_1 = -\theta_2 = +\pi$, which is again equal to

$$\mathcal{D} = 2\pi f_{\text{eff}} = \int_{-\pi}^{\pi} dt \sqrt{\kappa_{ij} \dot{\theta}^i \dot{\theta}^j} = 4\pi f \sqrt{1 - \frac{\epsilon}{2}}. \quad (6.11)$$

Let us see what happens when we apply our bound (4.21) to the instanton of charge $q_i = (1, 1)$ with action $S_{(1,1)} = S_1 + S_2 \geq 2$. This gives

$$\sqrt{\frac{2}{p}} \leq \|z\| \leq \frac{\sqrt{q_i \kappa^{ij} q_j}}{2} = \frac{1}{f \sqrt{2\epsilon}}. \quad (6.12)$$

Taking $\epsilon f^2 \rightarrow 0$, $f \rightarrow \infty$, we find that this bound can be satisfied even though f_{eff} can be parametrically large: the BPS axion WGC, applied to the instanton of charge $q_i = (1, 1)$, does not usefully restrict axion alignment.

To place a meaningful bound, we would instead need to consider the instanton of charge $q_i = (1, -1)$. This instanton has

$$\|z\| = \frac{\sqrt{q_i \kappa^{ij} q_j}}{S_{(1,-1)}} = \frac{2}{f_{\text{eff}} S_{(1,-1)}}. \quad (6.13)$$

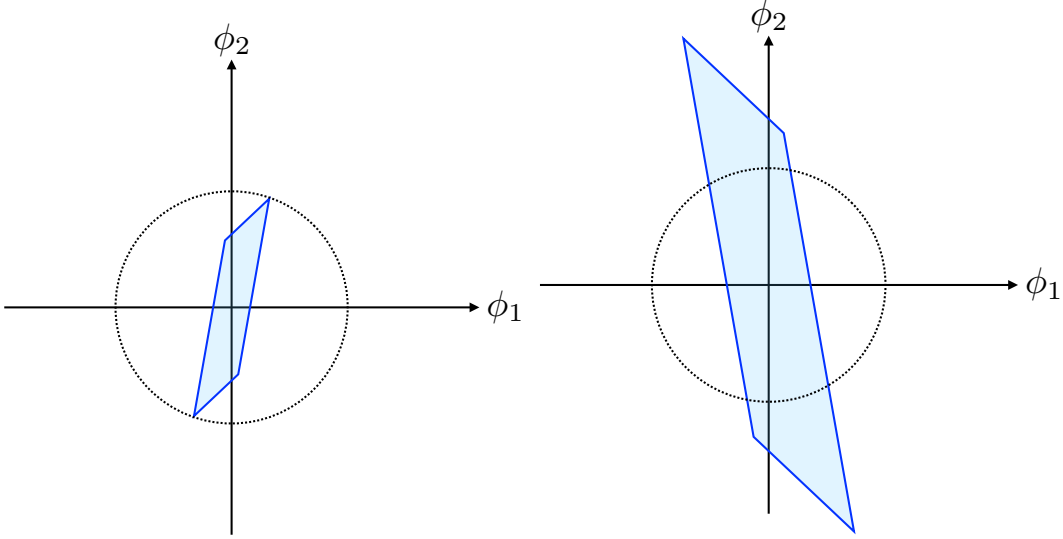


Figure 4: One orientation of alignment is incompatible with the BPS axion WGC (left), but the other is not (right). Here, the blue shaded region represents the axion fundamental domain, while the dashed black circle has radius $r = \pi\sqrt{p/2}$. The axes ϕ_1 and ϕ_2 depict canonically normalized axions.

From here, the BPS axion WGC would yield the familiar result $f_{\text{eff}} \leq \sqrt{p/2}$ if we could argue $S_{(1,-1)} \geq 2$, as we did in the previous example above. However, the charge $(1, -1)$ lies outside the BPS cone, so we cannot necessarily apply the BPS axion WGC to it, nor can we simply state that the instanton action is given by the sum of the instanton actions for the generators of the BPS cone.

We conclude that one orientation of kinetic alignment is ruled out by the BPS axion WGC, while the other is not. The difference between these two is depicted in Figure 4. This is related to the observation of [6], which rigorously ruled out decay constant “anti-alignment” of C_4 axions in Type IIB compactifications but could not rule out alignment in the opposite orientation.

This result highlights an important lesson: the approaches utilized in §4 and §5 of this paper do not imply a convex hull condition on instanton charge-to-action vectors, as they do not apply to instantons outside the BPS cone. In contrast, the approach of §3 does imply a convex hull condition in the situations (when it applies), as does the sharpened axion WGC (1.2). These bounds do rule out the models of kinetic alignment presented in this subsection.

However, it must be noted that none of the axion WGC bounds presented here—nor those derived in [26, 27]—have definitively ruled out natural inflation. The “extra

instanton loophole” [9, 92, 93, 98] and the “small action loophole” [99] remain open,⁹ and until they are closed or threaded by means of an explicit model, the question of natural inflation in quantum gravity cannot be answered definitively.

7 Discussion

In this work, we have used three different approaches to derive precise statements of the axion WGC in supersymmetric quantum gravity theories. The results obtained are complementary in the sense that whenever two or more approaches apply, they yield the same results. All of the bounds we obtain are consistent with—and slightly stronger than—the sharpened axion WGC bound (1.2) obtained in the recent works [26, 27].

It must be emphasized that the agreement between the three approaches was not guaranteed *a priori*. In particular, while our second and third approaches (involving homogeneous Kähler potentials and instanton α -vectors, respectively) dealt specifically with BPS instantons, the first approach instead used the string duality web in combination with higher-form versions of the WGC to establish a convex hull condition that (conjecturally) applies both inside and outside the cone of BPS instantons. The fact that our approaches yield identical bounds raises the interesting possibility that the bounds on BPS instantons derived in §4 and §5 can always be extended to convex hull conditions over the entirety of the instanton charge lattice. Further work in this direction is needed.

Our work suggests that BPS instantons in 4d supergravity theories must satisfy a bound of the form

$$f_q S_q \leq \sqrt{\frac{p}{2}} M_{\text{Pl}}, \quad (7.1)$$

where p is an integer between 1 and 7, inclusive. Within the string/M-theory landscape, there are several especially noteworthy values of p . 1/2-BPS instantons in maximal supergravity (as well as D(−1)-brane instantons charged under the universal axion C_0 in Type IIB compactifications) have $p = 1$ and saturate the bound (4.21). C_4 axions in O3/O7-orientifolds and C_2 axions in O5/O9 orientifolds have $p = 3$, which is also the value associated with instantons coming from particles in pure 5d supergravity wrapped on a circle. The largest value observed, $p = 7$, corresponds to C_3 axions in M-theory compactified on G_2 manifolds. In §4.6.2, we have given a suggestive (but not rigorous) argument that the imaginary distance bound may require the stronger condition $p \leq 3$ for any axion that is part of the modulus field determining a gauge coupling. Our work also suggests that BPS axion strings in 4d supergravity theories obey a bound of the

⁹See §7.1.5 of [23] for a review.

form

$$\frac{\mathcal{T}_{\tilde{q}}}{\|\tilde{q}\|} \leq \sqrt{\frac{p}{2}} M_{\text{Pl}}, \quad (7.2)$$

for the same value of p .

In this work, we have largely ignored the topic of gravitational instantons. However, it is interesting to note that in axion-dilaton-gravity, wormhole solutions do not exist above the critical value

$$\|\alpha\| = \sqrt{\frac{d-2}{d-1}}, \quad (7.3)$$

in d spacetime dimensions. This corresponds to the value $\|\alpha\| = \sqrt{2/3}$ in 4d, which in turn corresponds to the bound $p \leq 3$ on homogeneous Kähler potentials studied in §4. In contrast, for $\|\alpha\| < \sqrt{2/3}$, gravitational wormholes do not merely exist, but they also have a *larger* charge-to-action ratio $\|z\| = \|\alpha\|$ than do the BPS instantons, as observed in [50, 51, 100]. We still do not have an explanation for this strange phenomenon.

We observed a phenomenon of superpotential splitting, in which a single homogeneous term of the Kähler potential may split into a sum of multiple homogeneous terms in such a way that the sum of the degrees of homogeneity $p_{\text{tot}} = \sum_{\alpha} p_{\alpha}$ is preserved. This splitting manifested as a diagonal direction in the convex hull of α -vectors. It would be worthwhile to further explore this phenomenon and its consequences.

Finally, we note that while [26, 27] have placed the axion WGC on a sounder footing, they have also raised new questions and provided new tools that should be applied broadly. Is an instanton obeying the axion WGC the only cure for the pathologies associated with Giddings-Strominger wormholes, and if not, do the alternatives have interesting physical implications of their own? Are there other applications of the gravitational path integral to axion physics? These questions may shed further light on the Strong CP problem, dark matter, inflation, and dark energy.

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